EN 221: Intermediate Microeconomics analysis II

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Market structures

Market structure

- According to Bain the four main features of market structure;
  i. The degree of sellers concentration- This is the number and size distribution of firms producing a particular commodity or types of commodities in a market
  ii. The degree of buyers concentration. This shows the number and size distribution of buyers for the commodity in the market

Market structure…

- According to Bain the four main features of market structure;

  iii. The degree of product differentiation. Shows the difference in the products of different firm in the market

  iv. The condition of entry to market. This shows the relative ease with which new firms can join the category of sellers i.e firms in the market

Note:

- Each of the four dimensions/ features is important in determining the behavior of the firms which in turn will affect their performance as well as performance of the industry as a whole

Market Structures

	Perfect Competition	Monopolistic Competition	Oligopoly	Pure Monopoly
No. and Size of Firm	Many	Many	A Few	One
Extent of Product Differentiation	Identical	Different	Identical or Different	No Close Substitute
Barriers to Entry	None	None	Moderate to Difficult	Blocked

Market structures

- Perfect competition
- Monopoly
- Monopolistic competition
- Oligopoly (Game theory)
- Monopsony
- Non price competition

Market structures

- Nicholson, W and Synder C (2011). Microeconomics theory, basic principles and extensions, 11th edition, Fortworth, London, Dryden Press (Chapters 8,12,14 &15)
- Varian (2010) Intermediate Microeconomics. Chapters (24,25,27,28 &29

Price and output under perfect competition

Introduction

- The concept of **perfect competition** was first introduced by Adam Smith in his book "Wealth of Nations".
- **"*Perfect competition* is a market structure in which there are many firms selling identical products with no firm large enough, relative to the entire market, to be able to influence market price".**

Characteristics of PC Markets

- There are **large number** of buyers and sellers who compete among themselves and their number is so large that no buyer is in a position to influence the demand or supply in the market.

Characteristics of PC Markets…

- The commodity dealt in it is **homogenous** (goods produced by different firms are identical in nature)

Characteristics of PC…

- Every firm is **free** to enter the market or to go out of it
- There is a **perfect knowledge** on part of buyers and sellers.

Characteristics of PC…

- Facilities exists for movement of goods from one centre to another.
  - Also buyers have no preference as between different sellers and as between different units of commodity offered for sale
  - Sellers are quite indifferent as to whom they sell

- All firms are **price takers** they have to accept the price determined by market forces of demand and supply

Characteristics of PC…

- There is no government regulation
  - (**No government intervention** in the market – tariffs, subsidies, rationing of production etc are ruled out)
- Each producer supply a very small proportion of total industry output

Characteristics of PC…

- Consumers and producers have **perfect knowledge** about the market
- **Profit** maximization – the goal of all firms

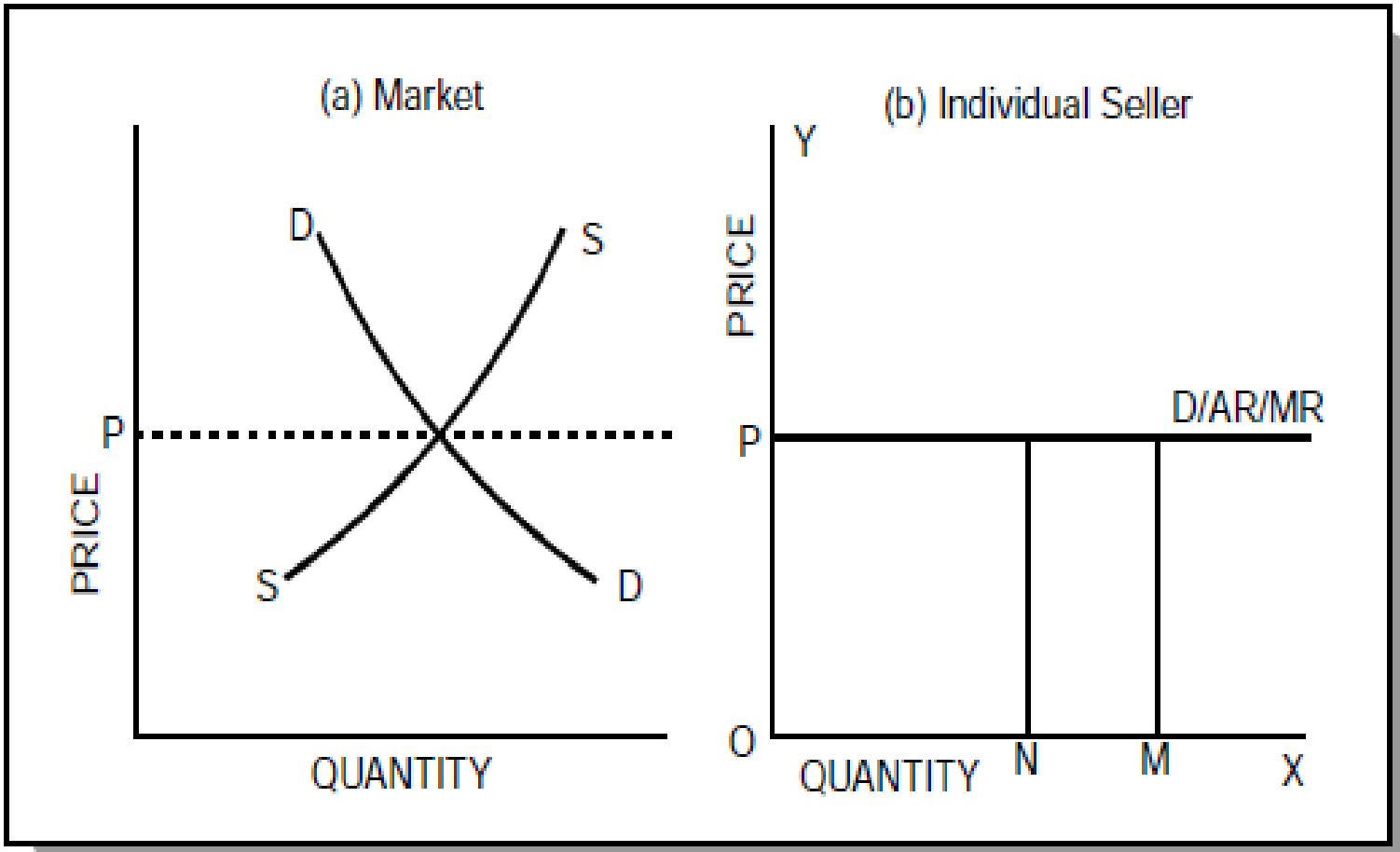
Firm demand under perfect competition (PC)

- The key condition for PC market is price taking
- Each firm and each consumer must take the market price of good as given

Firm demand under perfect competition (PC)

- No one can affect the price by their choice of how much to buy or sell.
- Individual firm will face a horizontal demand curve at the market price established by demand and supply

Fig: Firm's demand under perfect competitive market



- Industry price is fixed at P through interaction of demand and supply
- Firms have to accept price as given (price takers)
- P line acts as demand curve for the firm
- Firm have to choose the level of output that yield maximum profit

- Equilibrium price for the industry

Price ($)	Demand (units)	Supply (Units)
1	60	5
2	35	35
3	20	45
4	15	55
5	10	65

- Equilibrium price- at interaction between demand and supply (2 $)
- The individual firms will accept (2 $) per unit as price and sell different quantities at this price.
- Lets consider firm A firm A’s quantity sold, total revenue, average revenue and marginal revenue are given

Price ($)	Quantity sold	Total revenue	Average revenue	Marginal revenue
2	8	16	2	2
2	10	20	2	2
2	12	24	2	2
2	14	28	2	2
2	16	32	2	2

AR=MR=P

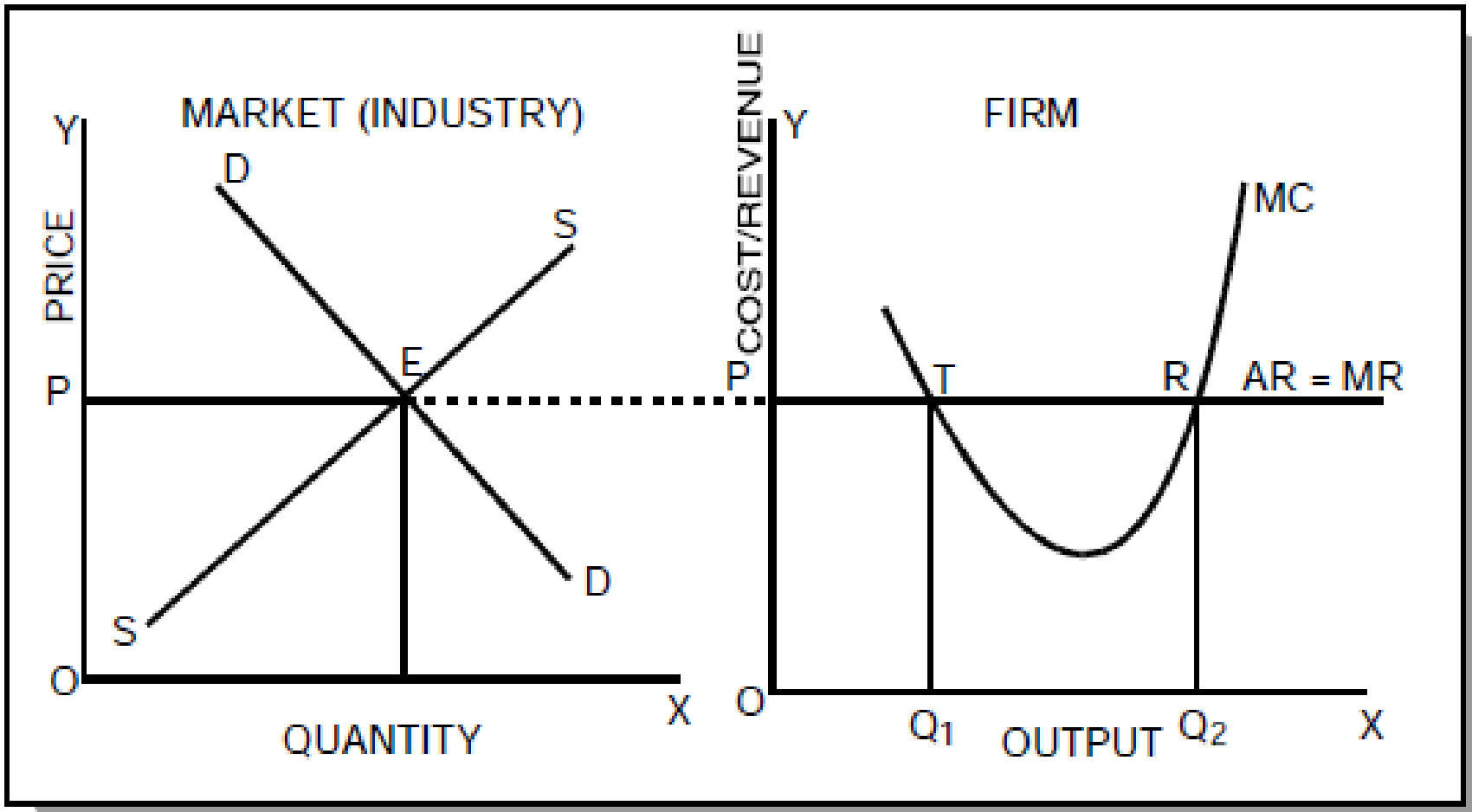
CONDITIONS FOR EQUILBRIUM OF A FIRM UNDER PC MARKET

- **The marginal revenue should be equal to marginal cost (MR = MC)**
- If MR>MC, there is an incentive for firm to expand its production further and gain by sale of additional units
- If MR<MC, the firm will have to reduce output since an additional unit adds more to costs than to revenue.

CONDITIONS FOR EQUILBRIUM OF A FIRM UNDER PC MARKET

- **The marginal revenue should be equal to marginal cost (MR = MC)**
- Thus: Profits are maximum only at point where MR=MC
- **The MC curve should cut MR curve from below (MC should have a positive slope)**

Equilibrium of the firm



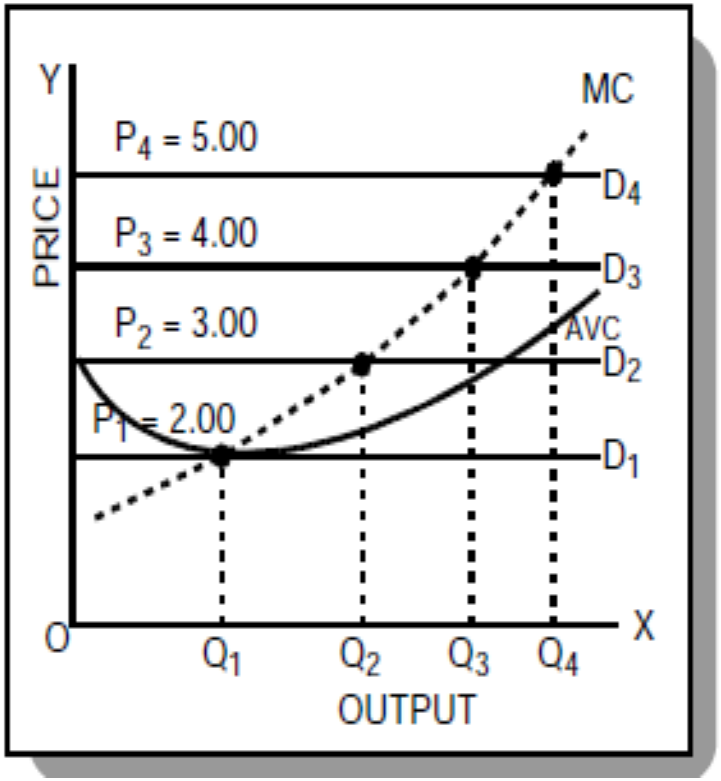
- In figure, DD and SS are industry demand and supply curves
- P in price/ and P line is demand curve (perfect elastic)
- MC cuts MR at two places T and R respectively
- T is not equilibrium as the second condition is not satisfied.
- 0Q is equilibrium level of output

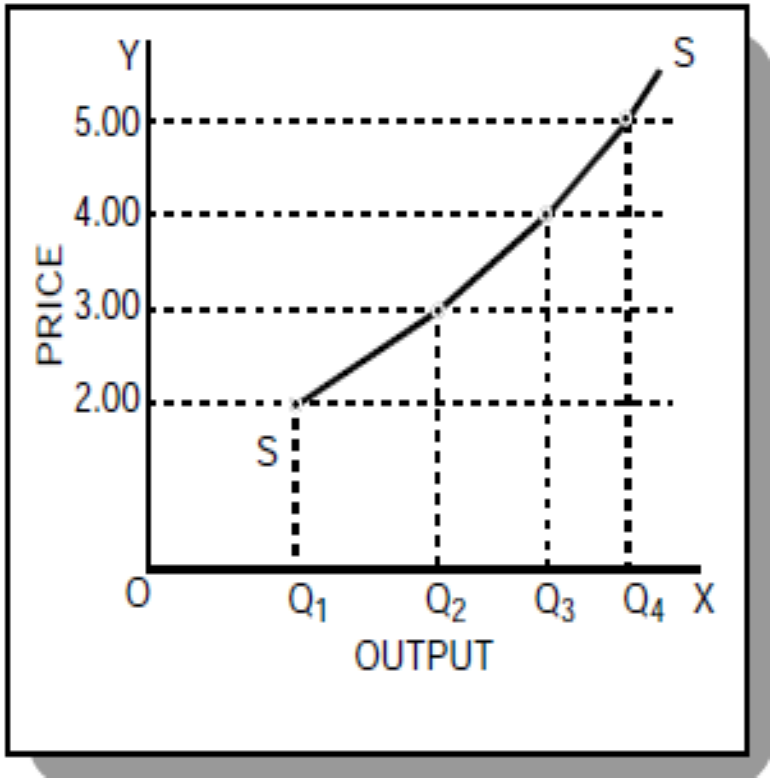
Market supply under perfect competition

- In order to determine the equilibrium of the industry we need to derive the market supply
- Market supply- is the sum of the supply of all the firms in the industry

- One interesting thing about MC curve of a firm under perfect competitive industry is that it depicts the firms supply curve.
- See example below

MC and supply curve for price taking firm





- The firm’s marginal cost curve which gives the marginal cost corresponding to each level of output is firms’ supply curve that gives various quantities the firm will supply at each price.
- For price below AVC, the firm will supply zero units because the firm is unable to meet even its variable costs

- For the prices above AVC, the firm will equate price and marginal cost
- When price is high enough to meet the AVC, the firm will decide to continue to continue its production.

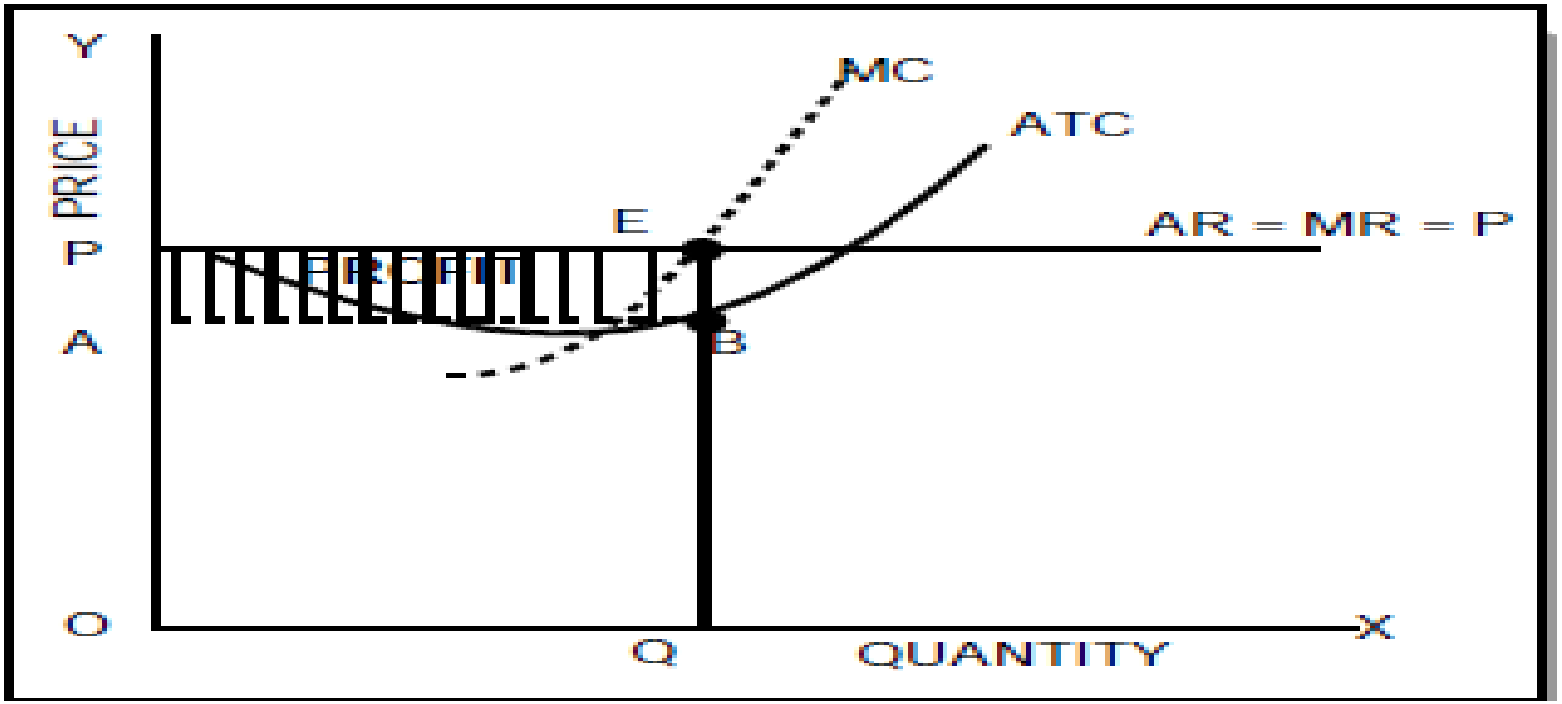
Can a competitive firm earn profits?

- In the short run, a firm will attain equilibrium position and at the same time it may earn
  - Supernormal profits
  - Normal profits or
  - Losses depending upon its costs conditions

- When average revenue of the firm = average total costs, the firm earn normal profits.
- For Supernormal profits average revenue > average total costs.

Short run equilibrium: Supernormal

- profit of a competitive firm



- Figure above shows how a firm can earn supernormal profits in the short run
- The diagram show in order to attain equilibrium, the firm tries to equate marginal revenue with marginal cost.
- MR curve is horizontal line and MC curve is U shaped curve which cuts MR at E

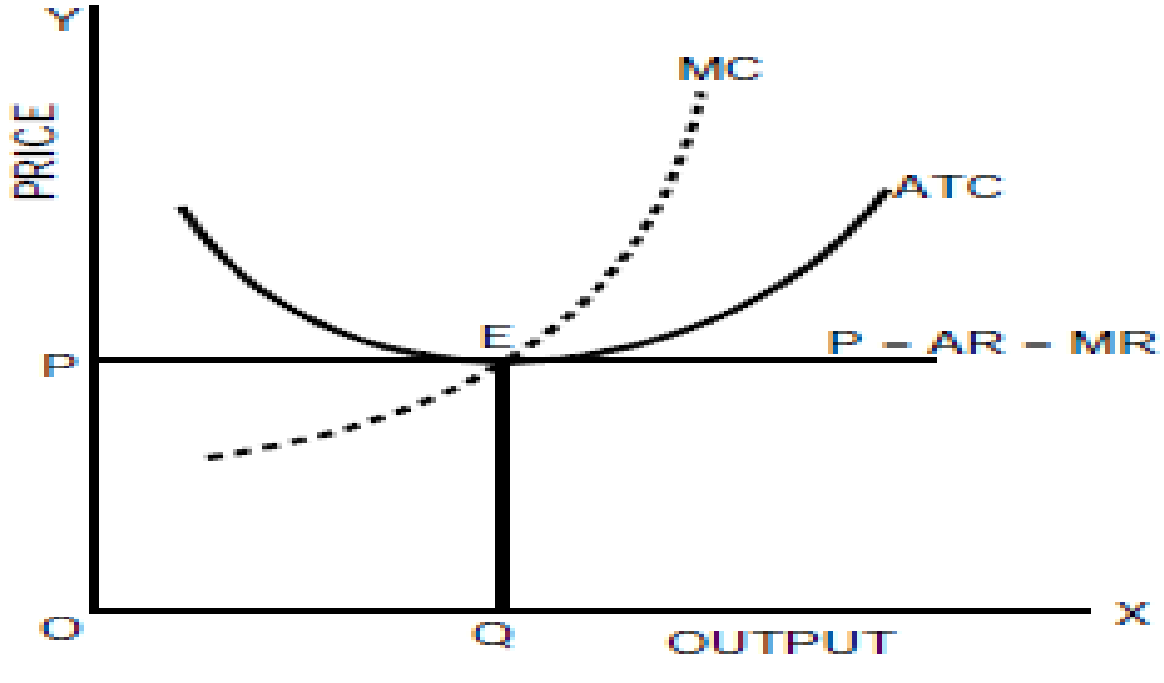
- At E, MR = MC. OQ is equilibrium output for the firm.
- The firm profit per unit is EB(AR-ATC), AR is EQ and ATC is BQ.
- Total profits are ABEP

Normal profits

- When firm just meets its average cost, it earns normal profit
- AR=ATC

Short run equilibrium of PC firm: normal

profits



- The figure shows that MR =MC at E
- The equilibrium output is OQ
- Since AR = ATC or OP = EQ, the firm is earning normal profit

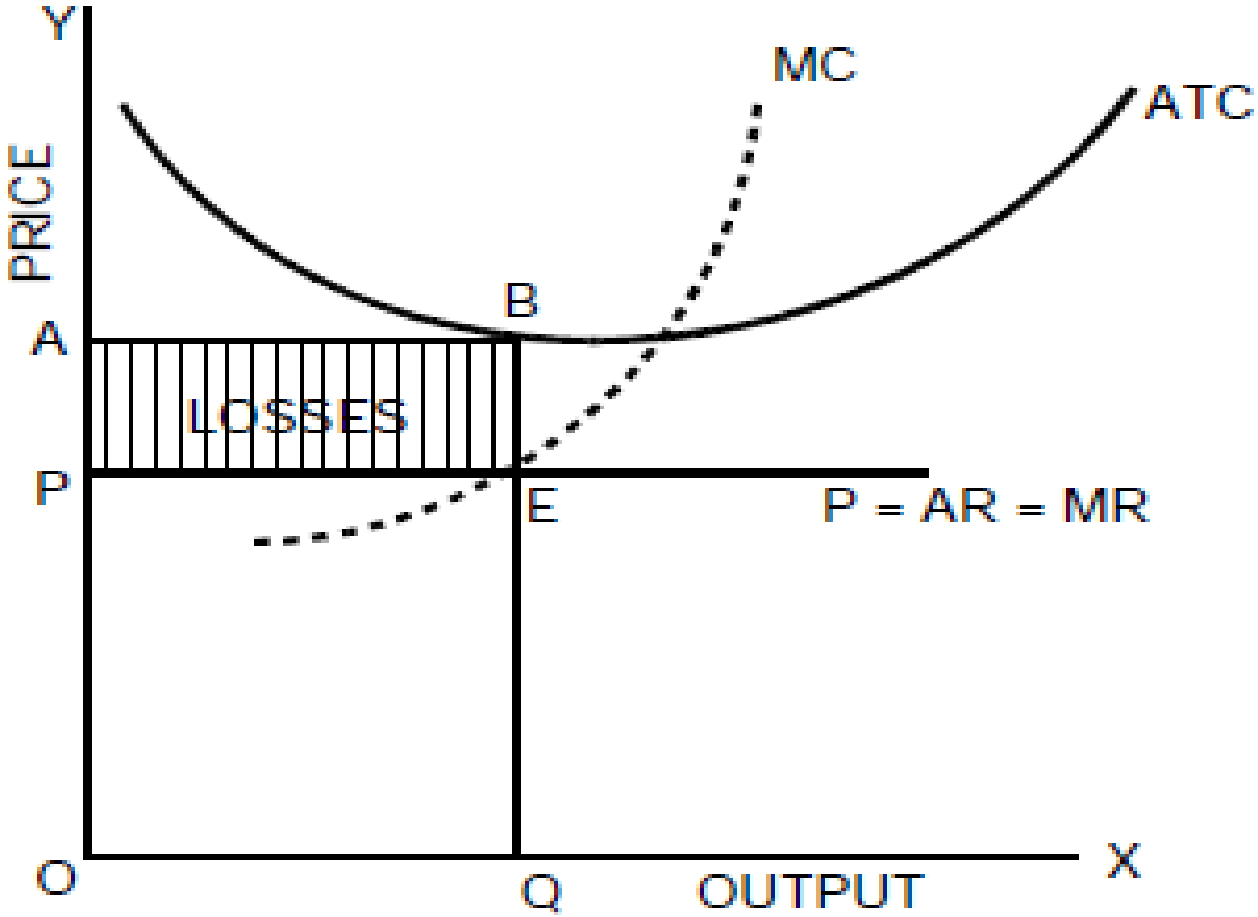
Losses

- The firm can be in equilibrium position and still make losses.
- This is the position when the firm is minimizing losses

Losses

- When the firm is able to meet its variable costs and a part of fixed cost it will continue with production in the short run.
- If the firm is unable to meet its AVC, it will better for it to shut down

Short run equilibrium of PC firm: Losses

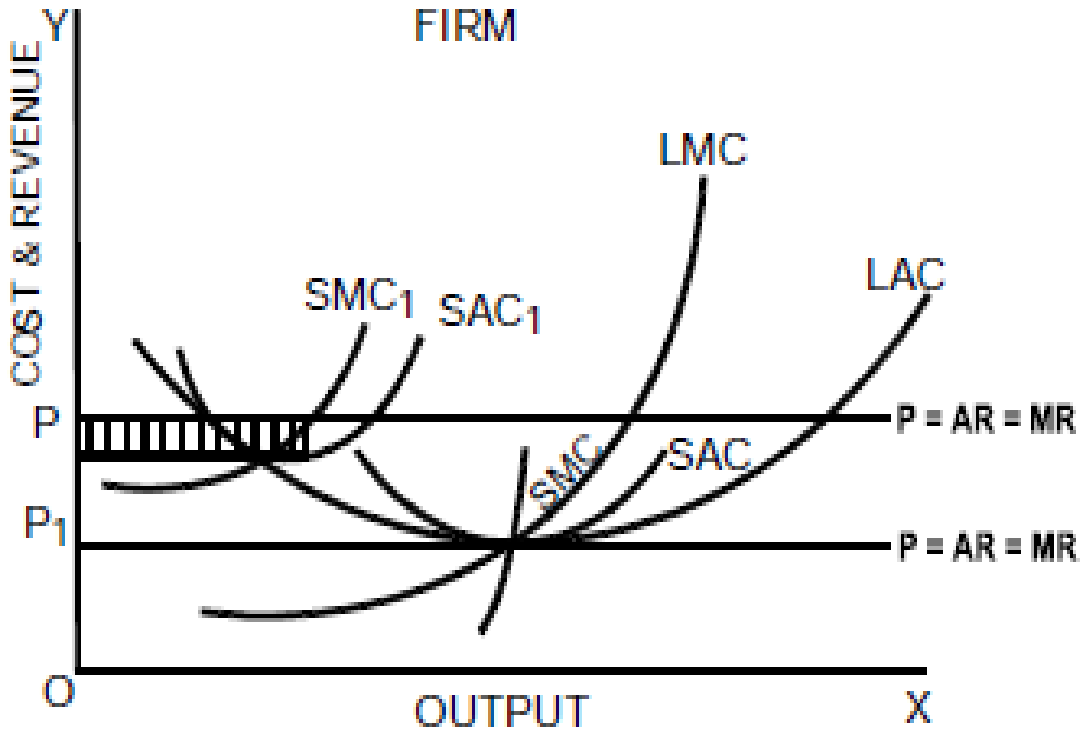


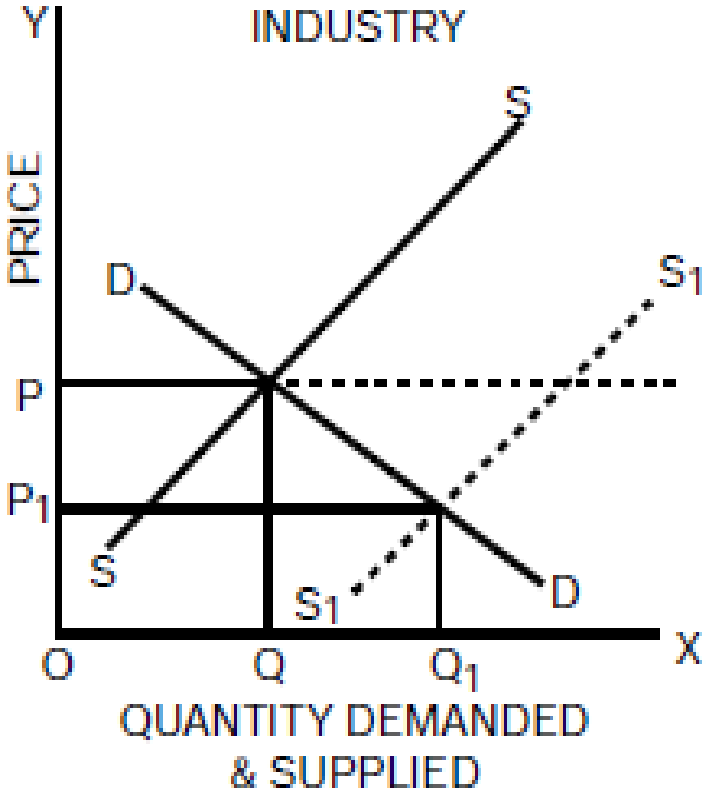
- In the figure, E is the equilibrium point and this point AR=EQ and ATC =BQ thus ATC>AR
- Since BQ>EQ, the firm is earning BE per unit loss and total loss is ABEP

- If they are making supernormal profits in short run – new firms will be attracted into the industry; this will lead to fall in price (downward shift in demand curves and upward shift in cost curves due to increase in price of factors as industry expands
- These changes will continue until ATC is tangent to demand curve

- If the firms make losses in the short run, they will leave the industry in the long run
- This will raise the price and costs may fall as the industry contracts, until the remaining firms in the industry cover their total costs inclusive of normal rate of profit

Long run equilibrium of a firm in a PC market





- The condition for long run equilibrium of a firm is that the marginal costs should be equal to the price and long run average costs
- LMC = LAC =P
- The firm adjusts its plant size so as to produce that level of output at which LAC is the minimum possible

- At the equilibrium the short run marginal cost is equal to long run marginal cost and short run average cost is equal to long run average cost.
- i.e in long run SMC=LMC=SAC=LAC=P=MR

Long run equilibrium for industry

- PC industry is in long run equilibrium when;

i. All the firms are earning normal profits only i.e all firms are in equilibrium

ii. There is no further entry or exit from the market

Working example

- In a market in the long run, each firm and potential entrants has a long run average cost curve $AC = 10Q^2 - 5Q + 20$

And long run marginal cost curve

$$MC = 30Q^2 - 10Q + 20$$

The market demand is given by

$$D(P) = 39000 - 2000P$$

i. At equilibrium, how many units will the firm produce?
ii. What is the market equilibrium price?
iii. What is the equilibrium number of firms in the long run?

- In perfectly competitive market the demand function is given by the equation $Q = 3000 - 6P$
- The firms operating in this market experience the cost function

$$TC = 4Q^2 + 12Q + 500$$

- Assume the market is in equilibrium

- Calculate the average cost curve and output in minimal average cost
- Find the supply curve of a single firm in the market
- How many firms operate on this market

Price and output under monopoly

Outline

- Introduction
- Profit maximization and output choice
- Monopoly and resource allocation
- Monopoly and product quality
- Price discrimination
- Regulation of monopolies

Introduction

- A monopoly is a single supplier to a market
- This firm may choose to produce at any point on the market demand curve

Why Monopoly exists?

- The reason a monopoly exists is that other firms find it unprofitable or impossible to enter the market
- Barriers to entry are the source of all monopoly power
  - there are two general types of barriers to entry
    - technical barriers
    - legal barriers

Technical Barriers to Entry

- The production of a good may exhibit decreasing marginal and average costs over a wide range of output levels
  - in this situation, relatively large-scale firms are low-cost producers
    - firms may find it profitable to drive others out of the industry by cutting prices
    - this situation is known as natural monopoly
    - once the monopoly is established, entry of new firms will be difficult

Technical Barriers to Entry

- Another technical basis of monopoly is <u>special knowledge of a low-cost productive technique</u>
  - it may be difficult to keep this knowledge out of the hands of other firms
- Ownership of unique resources may also be a lasting basis for maintaining a monopoly

Legal Barriers to Entry

- Many pure monopolies are created as a matter of law
  - with a patent, the basic technology for a product is assigned to one firm
  - the government may also award a firm an exclusive franchise to serve a market

Creation of Barriers to Entry

- Some barriers to entry result from actions taken by the firm
  - research and development of new products or technologies
  - purchase of unique resources
  - lobbying efforts to gain monopoly power
- The attempt by a monopolist to erect barriers to entry may involve real resource costs

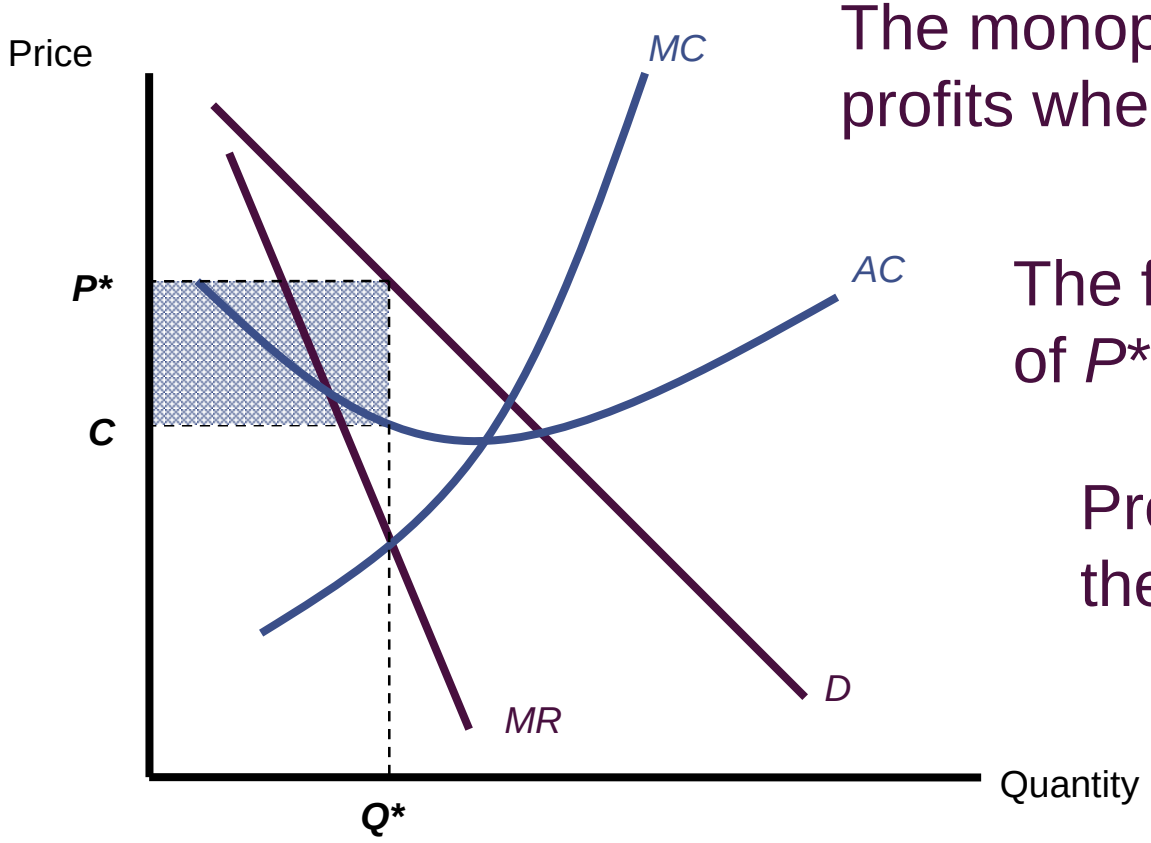
Profit Maximization and output choice

- To maximize profits, a monopolist will choose to produce that output level for which marginal revenue is equal to marginal cost
  - marginal revenue is less than price because the monopolist faces a downward-sloping demand curve
    - he must lower its price on all units to be sold if it is to generate the extra demand for this unit

Profit Maximization…

- Since $MR = MC$ at the profit-maximizing output and $P > MR$ for a monopolist, the monopolist will set a price greater than marginal cost

Profit Maximization…



The monopolist will maximize profits where *MR* = *MC*

The firm will charge a price of *P**

Profits can be found in the shaded rectangle

The Inverse Elasticity Rule

- The gap between a firm's price and its marginal cost is inversely related to the price elasticity of demand facing the firm

$$\frac{P - MC}{P} = \frac{1}{e_{Q,P}}$$

where $e_{Q,P}$ is the elasticity of demand for the entire market

The Inverse Elasticity Rule

- Two general conclusions about monopoly pricing can be drawn:
  - a monopoly will choose to operate only in regions where the market demand curve is elastic
    - $e_{Q,P} < -1$
  - the firm's "markup" over marginal cost depends inversely on the elasticity of market demand

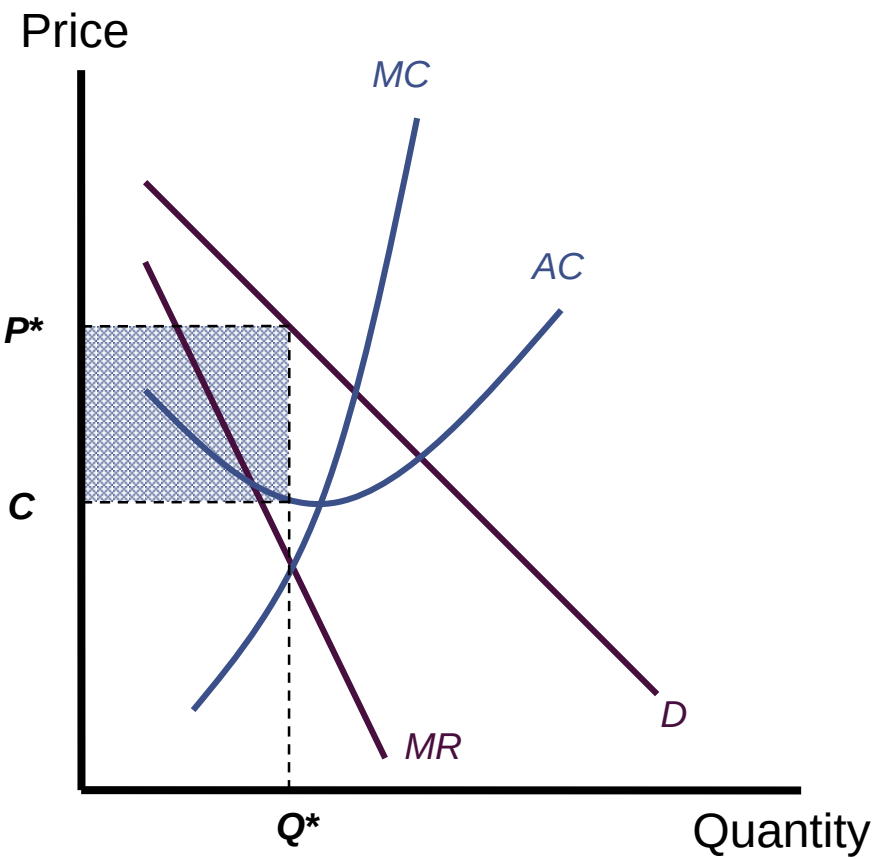
Monopoly Profits

- Monopoly profits will be positive as long as $P > AC$
- Monopoly profits can continue into the long run because entry is not possible
  - some economists refer to the profits that a monopoly earns in the long run as <u>monopoly rents</u>
    - the return to the factor that forms the basis of the monopoly

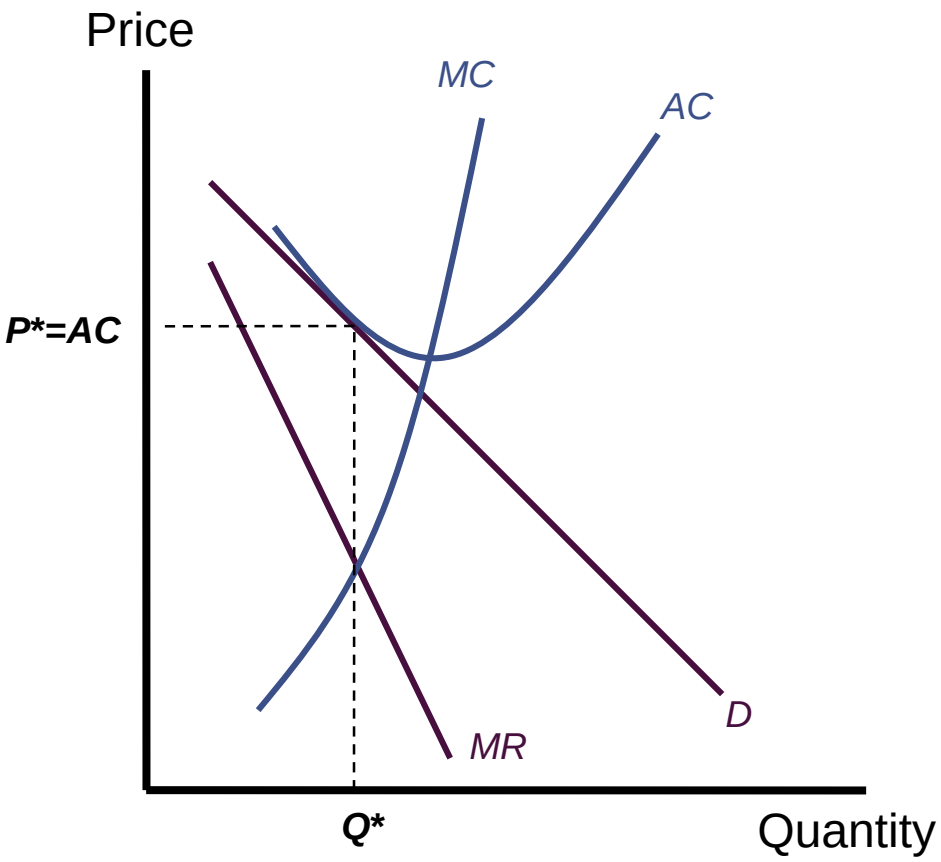
Monopoly Profits

- The size of monopoly profits in the long run will depend on the relationship between average costs and market demand for the product

Monopoly Profits



Positive profits



Zero profit

No Monopoly Supply Curve

- With a fixed market demand curve, the supply "curve" for a monopolist will only be one point
  - the price-output combination where *MR* = *MC*
- If the demand curve shifts, the marginal revenue curve shifts and a new profit-maximizing output will be chosen

Monopoly with Linear Demand

- Suppose that the market for frisbees has a linear demand curve of the form

$$Q = 2{,}000 - 20P$$

or

$$P = 100 - Q/20$$

- The total costs of the frisbee producer are given by

$$C(Q) = 0.05Q^2 + 10{,}000$$

Monopoly with Linear Demand

- To maximize profits, the monopolist chooses the output for which $MR = MC$
- We need to find total revenue

$$TR = P \cdot Q = 100Q - Q^2/20$$

- Therefore, marginal revenue is

$$MR = 100 - Q/10$$

while marginal cost is

$$MC = 0.1Q$$

Monopoly with Linear Demand

- Thus, $MR = MC$ where

$$100 - Q/10 = 0.1Q$$

$$Q^* = 500 \qquad P^* = 75$$

- At the profit-maximizing output,

$$C(Q) = 0.05(500)^2 + 10{,}000 = 22{,}500$$

$$AC = 22{,}500/500 = 45$$

$$\pi = (P^* - AC)Q = (75 - 45)\cdot 500 = 15{,}000$$

Monopoly with Linear Demand

- To see that the inverse elasticity rule holds, we can calculate the elasticity of demand at the monopoly's profit-maximizing level of output

$$\frac{dP}{dQ} \cdot \frac{Q}{P} = -2 \left(\frac{50}{50} \right)$$

Monopoly with Linear Demand

- The inverse elasticity rule specifies that

$$\frac{P - MC}{P} = \frac{1}{e_{Q,P}} = \frac{1}{3}$$

- Since $P^* = 75$ and $MC = 50$, this relationship holds

Practical example

1. A monopoly has demand curve which is given by equation Q=100-5P. Average cost curve is given by equation ATC=5Q. Calculate the firm's profit maximizing price and quantity. Prove that inverse elasticity rule hold
2. With the aid of graph and or mathematical notations differentiate between consumer surplus under monopoly from consumer surplus under Perfect competitive market

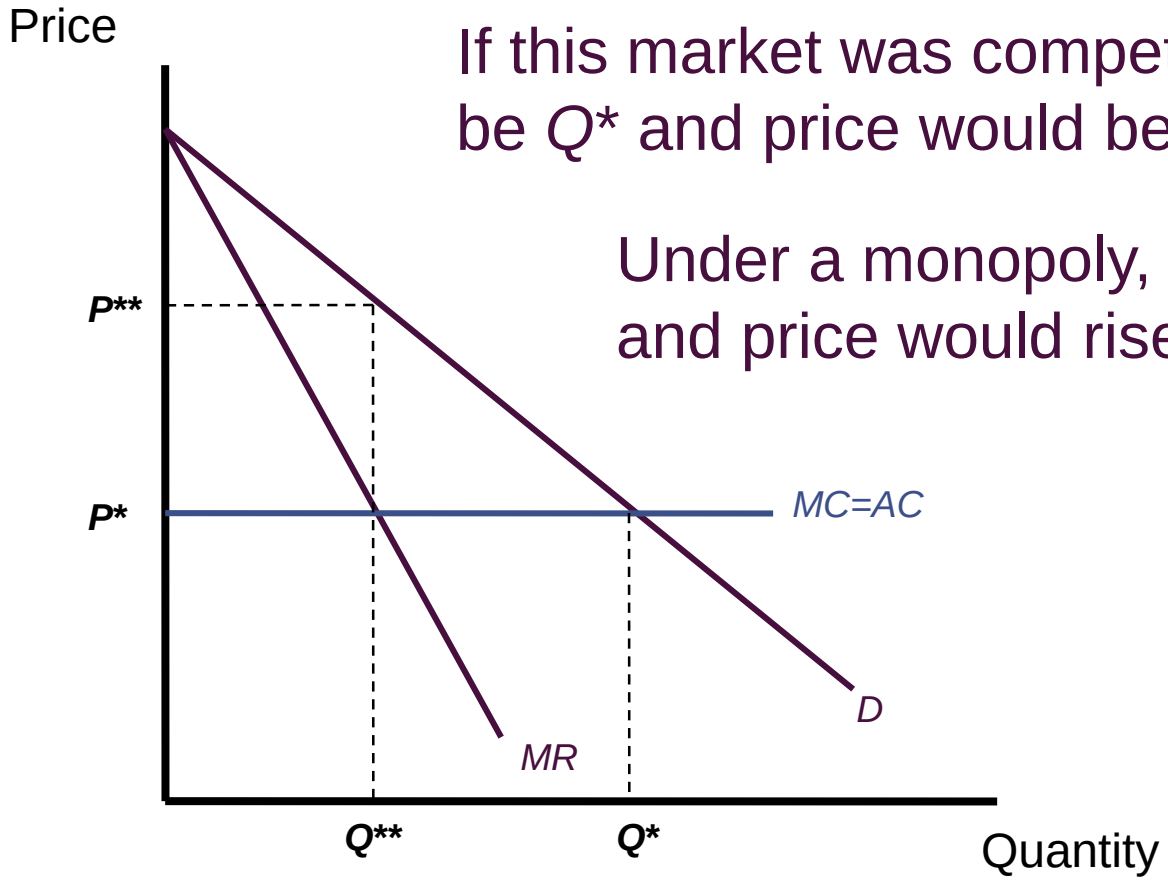
3. Given the demand of a firm in monopoly market $P=130 - 0.5Q$. Marginal cost $(MC)=2Q+10$ find consumer surplus, producer surplus and dead weight loss.

Deadline:

Monopoly and Resource Allocation

- To evaluate the allocational effect of a monopoly, we will use a perfectly competitive, constant-cost industry as a basis of comparison
  - the industry's long-run supply curve is infinitely elastic with a price equal to both marginal and average cost

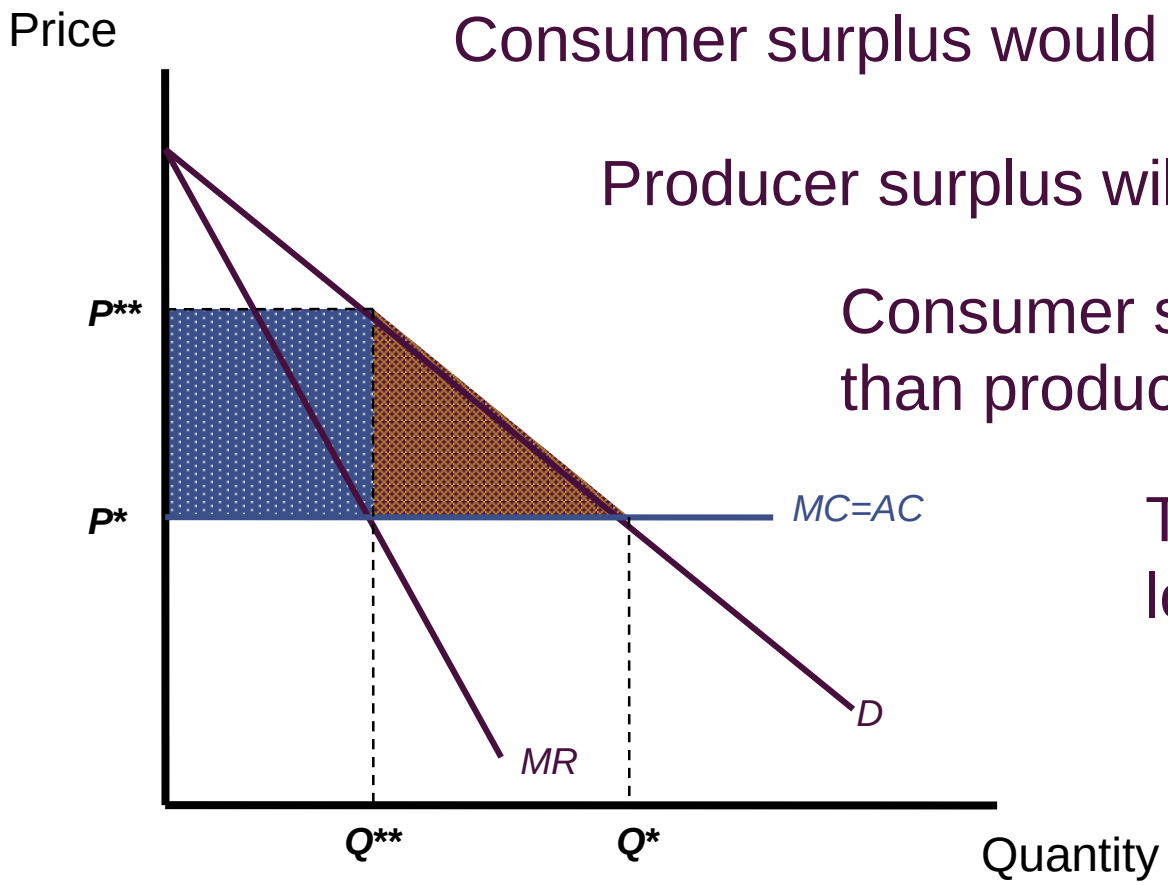
Monopoly and Resource Allocation



If this market was competitive, output would be *Q** and price would be *P**

Under a monopoly, output would be *Q*** and price would rise to *P***

Monopoly and Resource Allocation



Consumer surplus would fall

Producer surplus will rise

Consumer surplus falls by more than producer surplus rises.

There is a deadweight loss from monopoly

Welfare Losses and Elasticity

- Assume that the constant marginal (and average) costs for a monopolist are given by *c* and that the compensated demand curve has a constant elasticity:

$$Q = P^e$$

where *e* is the price elasticity of demand ($e < -1$)

Welfare Losses and Elasticity

- The competitive price in this market will be

$$P_c = c$$

and the monopoly price is given by

$$P_m = \frac{c}{1 + \frac{1}{e}}$$

Welfare Losses and Elasticity

- The consumer surplus associated with any price (P_0) can be computed as

$$CS = \int_{P_0}^{\infty} Q(P)\,dP = \int_{P_0}^{\infty} \frac{P^{e+1}}{e+1}\,dP$$

$$CS = \frac{P^{e+1}}{e+1}\Big|_{P_0}^{\infty} = -\frac{P_0^{e+1}}{e+1}$$

Welfare Losses and Elasticity

- Therefore, under perfect competition

$$CS_c = -\frac{c^{e+1}}{e+1}$$

and under monopoly

$$CS_m = -\frac{\left(\frac{c}{1+\frac{1}{e}}\right)^{e+1}}{e+1}$$

Welfare Losses and Elasticity

- Taking the ratio of these two surplus measures yields

$$\frac{CS_m}{CS_c} = \left(\frac{1}{1 + \frac{1}{e}} \right)^{e+1}$$

- If e = -2, this ratio is ½
  - consumer surplus under monopoly is half what it is under perfect competition

Welfare Losses and Elasticity

- Monopoly profits are given by

$$\pi_m = P_m Q_m - c_m Q_m = \left(\frac{c}{1-\frac{1}{e}} - c\right) Q_m$$

$$\pi_m = \left(\frac{c}{1-\frac{1}{e}} - c\right) \left(\frac{c}{1-\frac{1}{e}}\right)^{e} = \left(\frac{c}{1-\frac{1}{e}}\right)^{e+1} \cdot \frac{1}{e}$$

Welfare Losses and Elasticity

- To find the transfer from consumer surplus into monopoly profits we can divide monopoly profits by the competitive consumer surplus

$$\frac{\pi_m}{CS_c} = \frac{(e-1)\left(\frac{1}{1+\frac{1}{e}}\right)^{e-1}}{e} = \frac{(e-1)^{e}}{e^{e}}$$

- If e = -2, this ratio is ¼

Practical example

- Given the demand of a firm in monopoly market P=130 -0.5Q. Marginal cost (MC)=2Q+10 find consumer surplus, producer surplus and dead weight loss

- P=110, Q=40
- Cs = 400 Ps = area of trapezium 2400
- Dead weight loss =80

Monopoly and Product Quality

- The market power enjoyed by a monopoly may be exercised along dimensions other than the market price of its product
  - type, quality, or diversity of goods
- Whether a monopoly will produce a higher-quality or lower-quality good than would be produced under competition depends on demand and the firm's costs

Monopoly and Product Quality

- Suppose that consumers' willingness to pay for quality (X) is given by the inverse demand function $P(Q,X)$ where

$$\partial P/\partial Q < 0 \text{ and } \partial P/\partial X > 0$$

- If costs are given by $C(Q,X)$, the monopoly will choose Q and X to maximize

$$\pi = P(Q,X)Q - C(Q,X)$$

Monopoly and Product Quality

- First-order conditions for a maximum are

$$\frac{\partial \pi}{\partial Q} = P(Q) + Q\frac{\partial P}{\partial Q} - MC = 0$$

  – *MR* = *MC* for output decisions

$$\frac{\partial \pi}{\partial X} = Q\frac{\partial P}{\partial X} - MC_X = 0$$

  – Marginal revenue from increasing quality by one unit is equal to the marginal cost of making such an increase

Monopoly and Product Quality

- The level of product quality that will be opted for under competitive conditions is the one that maximizes net social welfare

$$S = \int_0^X P(x)\,dx - Q$$

- Maximizing with respect to *X* yields

$$\frac{\partial S}{\partial X} = P(X) - Q$$

Monopoly and Product Quality

- The difference between the quality choice of a competitive industry and the monopolist is:
  - the monopolist looks at the marginal valuation of one more unit of quality assuming that *Q* is at its profit-maximizing level
  - the competitve industry looks at the marginal value of quality averaged across all output levels

Monopoly and Product Quality

- Even if a monopoly and a perfectly competitive industry chose the same output level, they might opt for diffferent quality levels
  - each is concerned with a different margin in its decision making

Price Discrimination

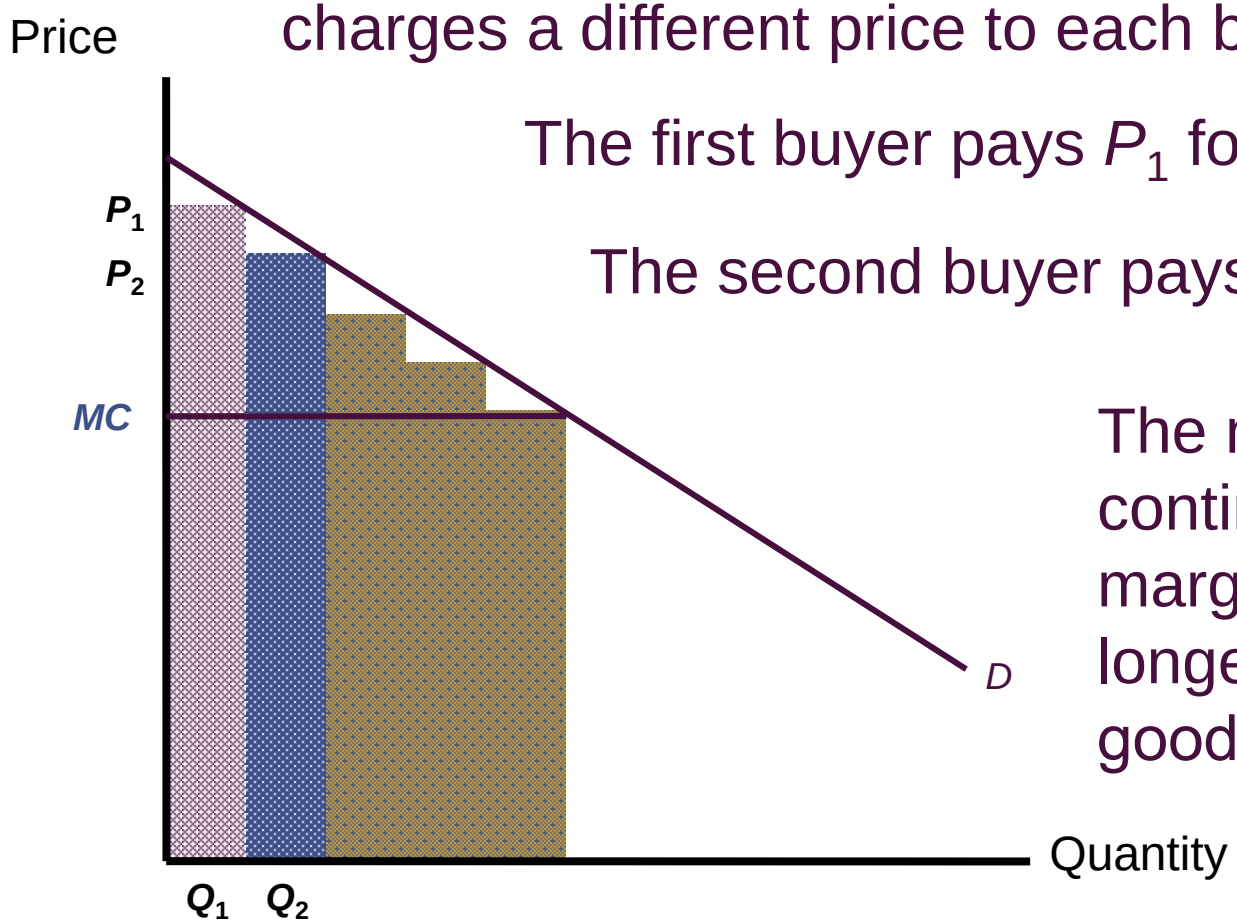
- A monopoly engages in price discrimination if it is able to sell otherwise identical units of output at different prices
- Whether a price discrimination strategy is feasible depends on the inability of buyers to practice arbitrage
  - profit-seeking middlemen will destroy any discriminatory pricing scheme if possible
    - price discrimination becomes possible if resale is costly

Perfect Price Discrimination

- If each buyer can be separately identified by the monopolist, it may be possible to charge each buyer the maximum price he would be willing to pay for the good
  - perfect or first-degree price discrimination
    - extracts all consumer surplus
    - no deadweight loss

Perfect Price Discrimination

Under perfect price discrimination, the monopolist charges a different price to each buyer



The first buyer pays P_1 for Q_1 units

The second buyer pays P_2 for Q_2-Q_1 units

The monopolist will continue this way until the marginal buyer is no longer willing to pay the good's marginal cost

Perfect Price Discrimination

- Recall the example of the frisbee manufacturer
- If this monopolist wishes to practice perfect price discrimination, he will want to produce the quantity for which the marginal buyer pays a price exactly equal to the marginal cost

Monopoly with Linear Demand

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$$MR = 100 - Q/10$$

while marginal cost is

$$MC = 0.1Q$$

Monopoly with Linear Demand

- Thus, $MR = MC$ where

$$100 - Q/10 = 0.1Q$$

$$Q^* = 500 \qquad P^* = 75$$

- At the profit-maximizing output,

$$C(Q) = 0.05(500)^2 + 10{,}000 = 22{,}500$$

$$AC = 22{,}500/500 = 45$$

$$\pi = (P^* - AC)Q = (75 - 45)\cdot 500 = 15{,}000$$

Perfect Price Discrimination

- Therefore,

$$P = 100 - Q/20 = MC = 0.1Q$$

$$Q^* = 666$$

- Total revenue and total costs will be

$$\text{Total revenue} = P \times Q = 40 \times 666 = \$26{,}640$$

$$\text{Total cost} = \int_0^{666} 0.1Q\,dQ = \$3{,}307$$

- Profit is much larger (23,333 > 15,000)

Market Separation

- Perfect price discrimination requires the monopolist to know the demand function for each potential buyer
- A less stringent requirement would be to assume that the monopoly can separate its buyers into a few identifiable markets
  - can follow a different pricing policy in each market
  - <u>third-degree price discrimination</u>

Market Separation

- All the monopolist needs to know in this case is the price elasticities of demand for each market
  - set price according to the inverse elasticity rule
- If the marginal cost is the same in all markets,

$$p_i\left(1-\frac{1}{\varepsilon_i}\right)=p_j\left(1-\frac{1}{\varepsilon_j}\right)$$

Market Separation

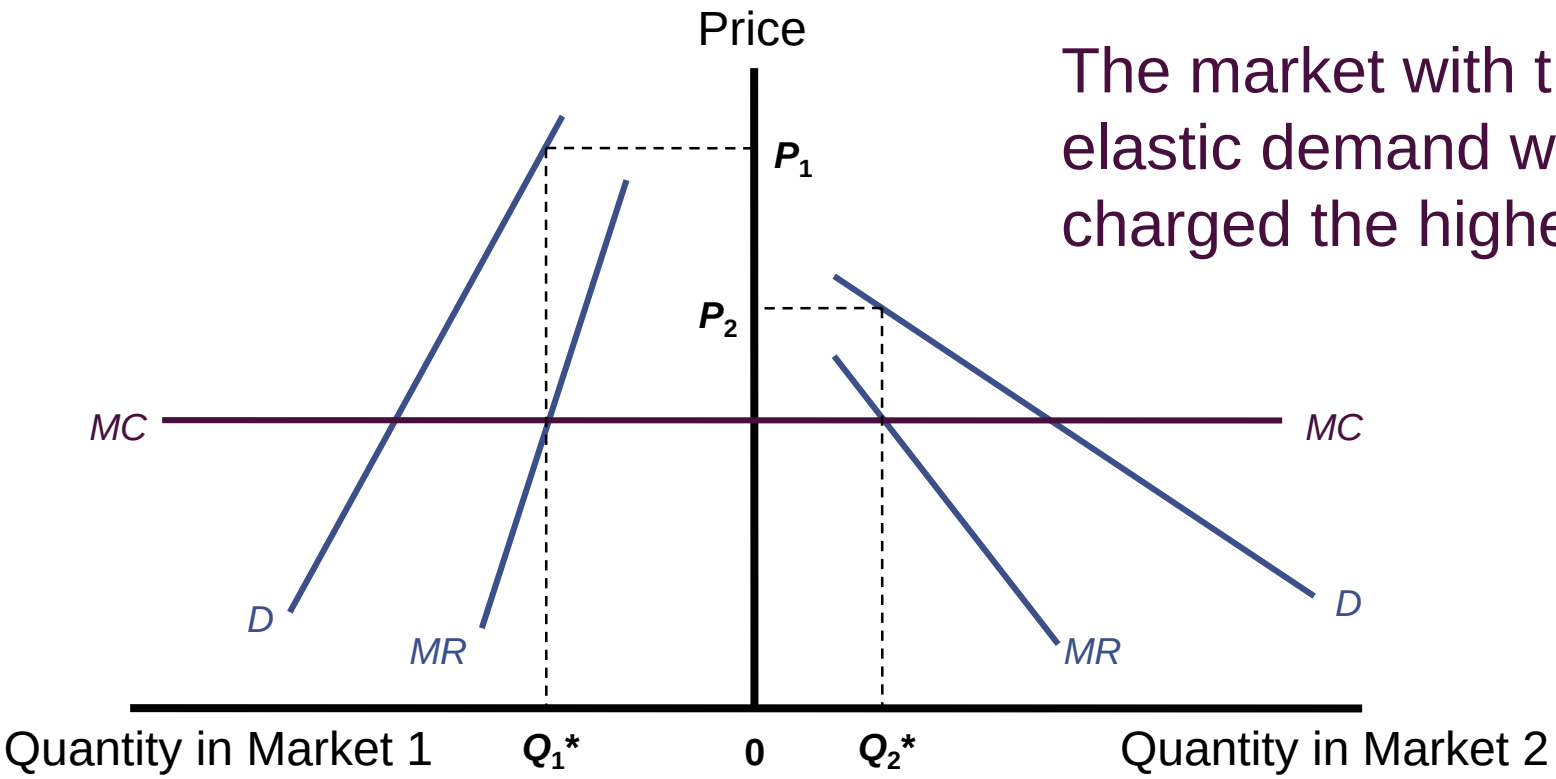
- This implies that

$$\frac{P_i}{P_j} = \frac{(1+\frac{1}{e_j})}{(1+\frac{1}{e_i})}$$

- The profit-maximizing price will be higher in markets where demand is less elastic

Market Separation

If two markets are separate, maximum profits occur by setting different prices in the two markets



The market with the less elastic demand will be charged the higher price

Third-Degree Price Discrimination

- Suppose that the demand curves in two separated markets are given by

$$Q_1 = 24 - P_1$$

$$Q_2 = 24 - 2P_2$$

- Suppose that $MC = 6$
- Profit maximization requires that

$$MR_1 = 24 - 2Q_1 = 6 = MR_2 = 12 - Q_2$$

Third-Degree Price Discrimination

- Optimal choices and prices are

$$Q_1 = 9 \qquad P_1 = 15$$

$$Q_2 = 6 \qquad P_2 = 9$$

- Profits for the monopoly are

$$\pi = (P_1 - 6)Q_1 + (P_2 - 6)Q_2 = 81 + 18 = 99$$

Third-Degree Price Discrimination

- The allocational impact of this policy can be evaluated by calculating the deadweight losses in the two markets
  - the competitive output would be 18 in market 1 and 12 in market 2

$$DW_1 = 0.5(P_1\text{-}MC)(18\text{-}Q_1) = 0.5(15\text{-}6)(18\text{-}9) = 40.5$$

$$DW_2 = 0.5(P_2\text{-}MC)(12\text{-}Q_2) = 0.5(9\text{-}6)(12\text{-}6) = 9$$

Third-Degree Price Discrimination

- If this monopoly was to pursue a single-price policy, it would use the demand function

$$Q = Q_1 + Q_2 = 48 - 3P$$

- So marginal revenue would be

$$MR = 16 - 2Q/3$$

- Profit-maximization occurs where

$$Q = 15 \qquad P = 11$$

Third-Degree Price Discrimination

- The deadweight loss is smaller with one price than with two:

$$DW = 0.5(P\text{-}MC)(30\text{-}Q) = 0.5(11\text{-}6)(15) = 37.5$$

Regulation of Monopoly

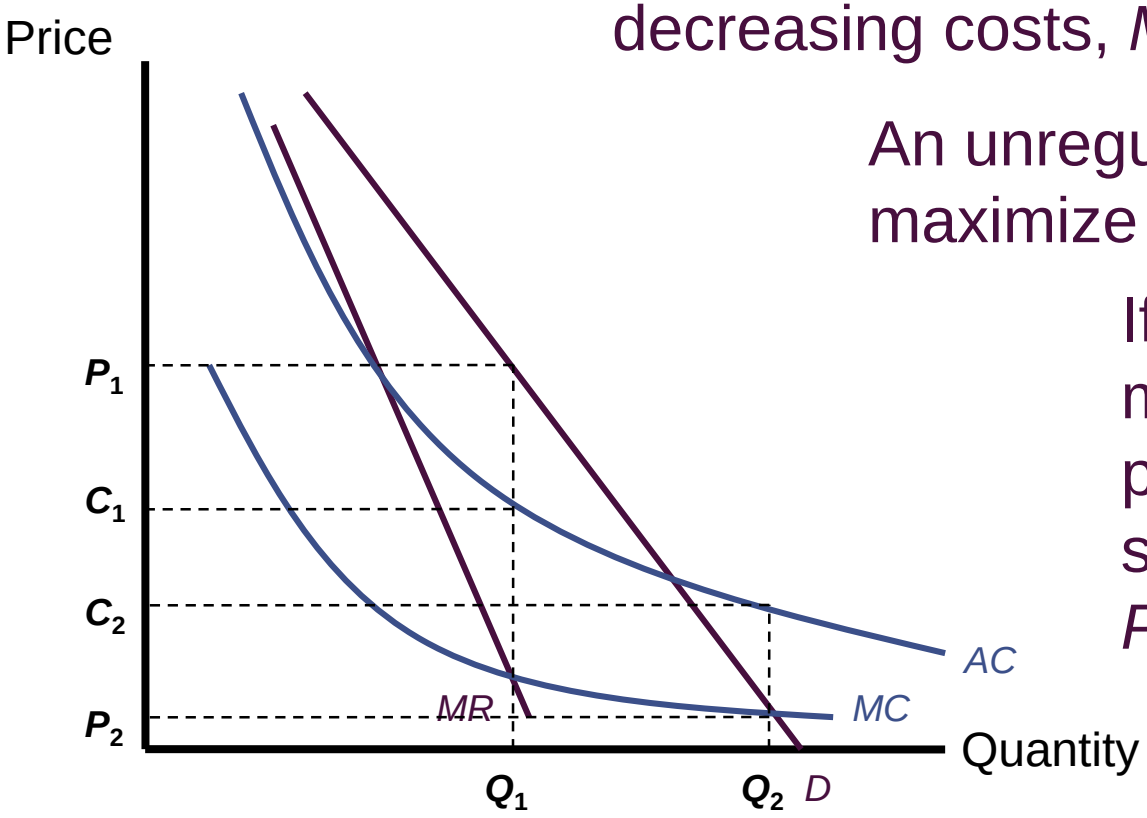
- Natural monopolies such as the utility, communications, and transportation industries are highly regulated in many countries

Regulation of Monopoly

- Many economists believe that it is important for the prices of regulated monopolies to reflect marginal costs of production accurately
- An enforced policy of marginal cost pricing will cause a natural monopoly to operate at a loss
  - natural monopolies exhibit declining average costs over a wide range of output

Regulation of Monopoly

Because natural monopolies exhibit decreasing costs, *MC* falls below *AC*

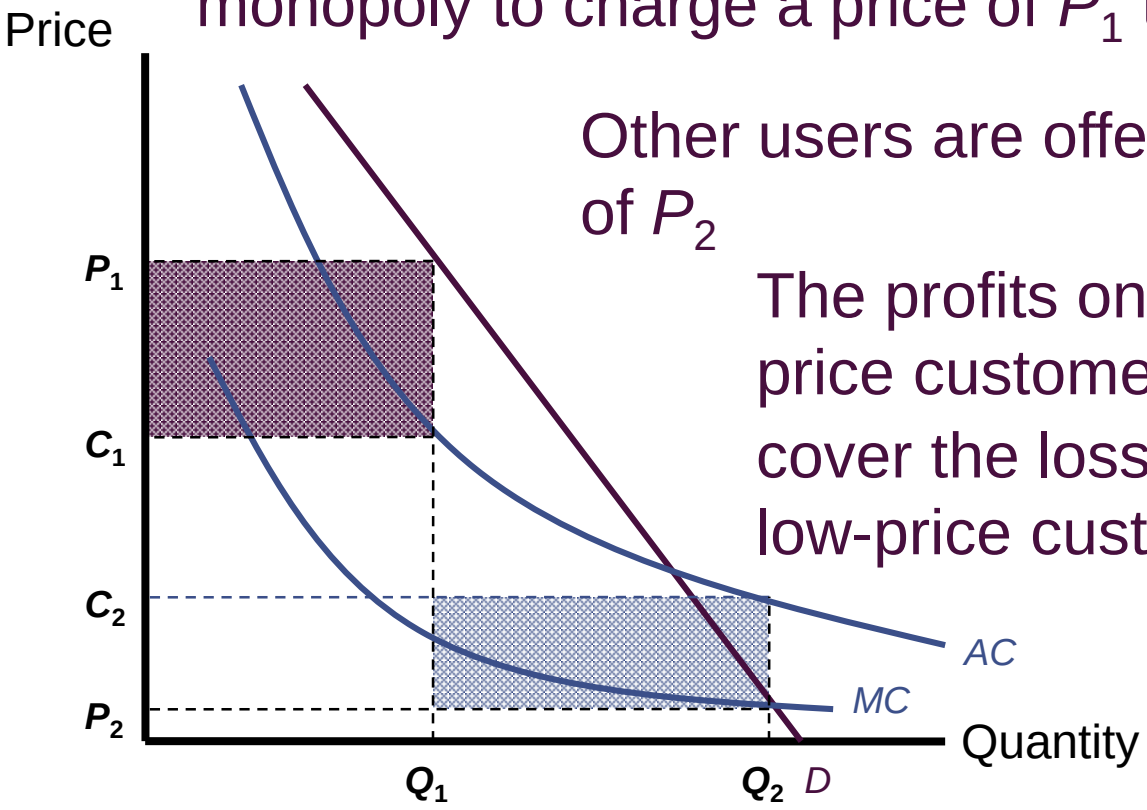


An unregulated monopoly will maximize profit at Q_1 and P_1

If regulators force the monopoly to charge a price of P_2, the firm will suffer a loss because $P_2 < C_2$

Regulation of Monopoly

Suppose that the regulatory commission allows the monopoly to charge a price of P_1 to some users



Other users are offered the lower price of P_2

The profits on the sales to high-price customers are enough to cover the losses on the sales to low-price customers

Regulation of Monopoly

- Another approach followed in many regulatory situations is to allow the monopoly to charge a price above marginal cost that is sufficient to earn a "fair" rate of return on investment
  - if this rate of return is greater than that which would occur in a competitive market, there is an incentive to use relatively more capital than would truly minimize costs

Regulation of Monopoly

- Suppose that a regulated utility has a production function of the form

$$q = f(k,l)$$

- The firm's actual rate of return on capital is defined as

$$s = \frac{p(f(k,l)) - wl}{k}$$

Regulation of Monopoly

- Suppose that s is constrained by regulation to be equal to s_0, then the firm's problem is to maximize profits

$$\pi = pf(k,l) - wl - vk$$

subject to this constraint

- The Lagrangian for this problem is

$$\mathbf{L} = pf(k,l) - wl - vk + \lambda[wl + s_0k - pf(k,l)]$$

Regulation of Monopoly

- If λ=0, regulation is ineffective and the monopoly behaves like any profit-maximizing firm
- If λ=1, the Lagrangian reduces to

$$\mathbf{L} = (s_0 - v)k$$

which (assuming $s_0 > v$), will mean that the monopoly will hire infinite amounts of capital – an implausible result

Regulation of Monopoly

- Therefore, $0<\lambda<1$ and the first-order conditions for a maximum are:

$$\frac{\partial L}{\partial p} = p f'(q)(1-\lambda) + \lambda$$

$$\frac{\partial L}{\partial k} = p f'(q)(1-\lambda) - \lambda$$

$$\frac{\partial L}{\partial \lambda} = \lambda \ln(p) - \epsilon$$

Regulation of Monopoly

- Because $s_0 > v$ and $\lambda < 1$, this means that

$$pf_k < v$$

- The firm will hire more capital than it would under unregulated conditions
  - it will also achieve a lower marginal productivity of capital

Dynamic Views of Monopoly

- Some economists have stressed the beneficial role that monopoly profits can play in the process of economic development
  - these profits provide funds that can be invested in research and development
  - the possibility of attaining or maintaining a monopoly position provides an incentive to keep one step ahead of potential competitors

Important Points to Note:

- The most profitable level of output for the monopolist is the one for which marginal revenue is equal to marginal cost
  - at this output level, price will exceed marginal cost
  - the profitability of the monopolist will depend on the relationship between price and average cost

Important Points to Note:

- Relative to perfect competition, monopoly involves a loss of consumer surplus for demanders
  - some of this is transferred into monopoly profits, whereas some of the loss in consumer surplus represents a deadweight loss of overall economic welfare
  - it is a sign of Pareto inefficiency

Important Points to Note:

- Monopolies may opt for different levels of quality than would perfectly competitive firms
- Durable good monopolists may be constrained by markets for used goods

Important Points to Note:

- A monopoly may be able to increase its profits further through price discrimination – charging different prices to different categories of buyers
  - the ability of the monopoly to practice price discrimination depends on its ability to prevent arbitrage among buyers

Important Points to Note:

- Governments often choose to regulate natural monopolies (firms with diminishing average costs over a broad range of output levels)
  - the type of regulatory mechanisms adopted can affect the behavior of the regulated firm